ModelIt!

Student Activity Pack

Can Sound Travel Through Space?

Why Astronauts Need Radios and Explosions Are Silent in a Vacuum

Grade 04L2 | NGSS 4-PS4-1, 4-PS4-3 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Do you think you could hear an explosion in outer space? Why or why not?

2 Why do you think you can hear someone talking underwater in a swimming pool?

3 Draw what you think is happening in the air between a drum being hit and your ear hearing the sound.

4 If sound needs something to travel through, what do you think happens in a place with nothing in it?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Vibration Strength | Medium Density | Wave Speed | Sound Loudness | Wave Absorption

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sound in Air

Vibration Strength at 70%, Medium Density at 30%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Sound in Water

Vibration Strength at 70%, Medium Density at 70%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Sound in Space

Medium Density at 0%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Distance from Source — How far the listener is from the sound — sound gets quieter over distance as the wave spreads out

Temperature — How hot or cold the medium is — warmer air carries sound slightly faster because molecules move more quickly

Frequency — How fast the vibrations happen — this determines whether we hear a high pitch or a low pitch

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

In every space movie, explosions make huge booming sounds. But is that what would really happen in space? Use evidence from your simulation to support your answers.

1 Why do space movies get the sound of explosions wrong?

2 If you were on the Moon with a friend standing right next to you, could you talk to each other without a radio? Why or why not?

3 Why does sound travel faster through water than through air?

4 How is sound traveling through a medium like a line of dominoes falling?

5 How did your model help you understand something you cannot directly experience, like silence in space?

8 STEM Challenge: Design the Best Sound Transmitter

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YOUR ENGINEERING MISSION

Design and test a system that transmits sound as clearly as possible from one side of the classroom to the other using different materials.

THE SCENARIO

A submarine crew needs to communicate between sealed rooms without using electronics. Your engineering team must design a sound transmission system that carries voices as clearly as possible through solid materials. Test different materials to find which one transmits sound the best and explain WHY using your model evidence.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU TEST AS YOUR SOUND TRANSMISSION MEDIUM (STRING, WIRE, WOOD, METAL, PLASTIC)?
2. HOW WILL YOU MEASURE WHICH MATERIAL TRANSMITS SOUND THE CLEAREST?
3. HOW WILL YOU KEEP YOUR TEST FAIR — SAME SOUND SOURCE, SAME DISTANCE, SAME LISTENER?
4. HOW WILL YOU CONNECT YOUR MATERIAL BETWEEN THE SENDER AND RECEIVER (CUPS, FUNNELS, DIRECT CONTACT)?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Acoustical Engineers design spaces and systems for optimal sound — from concert halls to noise-canceling headphones to submarine sonar systems. They earn $65,000–$110,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 An astronaut in space watches a satellite explode 50 meters away. What does the astronaut hear?

- **A.** A very loud boom because explosions are always loud
- **B.** A quiet boom because the sound gets weaker in space
- **C.** Nothing at all, because sound cannot travel through the vacuum of space
- **D.** A delayed boom because sound travels slowly in space

2 In the model, Vibration Strength and Medium Density are external components. What makes them external?

- **A.** They are the loudest parts of the model
- **B.** They are conditions we can set or control, and the other components respond to them
- **C.** They happen outside of the classroom
- **D.** They are the parts of the model that do not change

3 The model shows that when Medium Density increases (from air to water to steel), Wave Speed also increases. Which best explains why?

- **A.** Denser materials are heavier, so they push sound waves forward with more force
- **B.** In denser materials, molecules are packed closer together, so each molecule bumps into the next one faster
- **C.** Sound creates its own molecules in denser materials
- **D.** Denser materials have less friction, so sound slides through them easily

4 The relationship between Wave Absorption and Sound Loudness is NEGATIVE. What does this mean for how loud a sound is when it reaches you?

- **A.** When absorption increases, loudness also increases because the medium amplifies the sound
- **B.** Absorption and loudness always stay the same no matter what
- **C.** When absorption increases, loudness decreases because more sound energy is soaked up before reaching the listener
- **D.** Absorption only affects pitch, not loudness

5 A student sets Medium Density to 0% in the model, representing the vacuum of space. What happens to ALL the internal components?

- **A.** Wave Speed increases because there is nothing to slow the sound down
- **B.** Sound Loudness increases because there is nothing to absorb the sound
- **C.** All internal components drop to zero because there are no molecules to carry the sound wave
- **D.** Wave Speed stays the same but Sound Loudness decreases slightly